

Toby Tran

+61 411 739 097 | ngtong1406@gmail.com | linkedin.com/in/tobytran | github.com/ngtong1406

EDUCATION

University of South Australia

Bachelor of Information Technology (Software Development)

Mawson Lakes, SA

Mar. 2023 – Nov. 2025

- **GPA:** 6.17 / 7.0
- **Honours/Awards:** 2023-25 University Merit Award(s)
- **Relevant Coursework:** Data Structures Essentials, Object-Oriented Programming (OOP), Design Patterns with C++, Web Technology, Agile Development and Governance, Operating Systems and Tool Chains.

PROJECTS

Mission Planner | *FastAPI, PyWebView, ReactJS, TailwindCSS, CesiumJS, DBScan*

Jul. 2025 – Nov. 2025

- Developed a full-stack desktop application for an orbital debris removal startup (**Paladin Space**) to identify high-risk collision targets for commercial satellites, significantly reducing mission fuel costs.
- Applied a **layered backend architecture** that splits the system into multiple domains, combined with design patterns (*Strategy, Adapter, Factory*) for easier maintenance and future extensions.
- Implemented **10+ asynchronous FastAPI endpoints** for faster and more responsive data processing, with a **custom logging system** to facilitate debugging.
- Led data processing pipeline in the backend by crafting fetch requests based on the original API source and writing a Python script to transform fetched data into usable objects.
- Built an **automatic synchroniser** that uses the above Python script and internal logs to handle bulk data downloads, serialisation, and synchronisation (for up to **100,000+ CSV records**) in real-time without lagging.
- Integrated a 3D globe using ReactJS and CesiumJS to visualise the processed data by converting them into CZML-formatted objects.

SA Expiation Dashboard | *ASP.NET Core Web API, LINQ, ReactJS, Bootstrap, D3.js*

Jul. 2024 – Nov. 2024

- Developed a ReactJS dashboard with up to **6 different search/filter controls** to identify **optimal Mobile Phone Detection Camera (MPDC) set-up locations** from expiation data.
- Integrated secure authentication and login features with **SHA-256** password hashing.
- Implemented **6 WebAPI endpoints** to query location-based expiation records for strategic offense analytics, using LINQ (EF Core) queries to safely fetch MS-SQL data.
- Designed a navigable Bootstrap-styled template aligned with SAPOL branding.
- Visualised MPDC location insights from search results via **interactive D3.js graphs** incorporated within a detailed report page.

Towny Title | *Spigot API, Java, Maven, Git*

Oct. 2017

- Developed an open-source Minecraft server plugin to enhance player navigation and territory awareness for the Towny framework.
- Implemented **real-time player location tracking and event-driven triggers** to render dynamic on-screen titles and warnings with minimal server latency.
- Implemented cross-version compatibility across **6 major Minecraft releases** (v1.7 to v1.12) to support diverse server environments.
- Published the plugin to SpigotMC, reaching **1,200+ downloads** within the community.

TECHNICAL SKILLS

Languages: Java, Python, C#, C/C++, SQL (MySQL), JavaScript, HTML/CSS

Frameworks: ASP.NET Core (MVC, WebAPI), React, TailwindCSS, Node.js, FastAPI, Framer Motion

Developer Tools: Git, Docker, VS Code, Visual Studio, IntelliJ, Eclipse, Jira Atlassian (Scrum-based)